

VTU SportsOS

A Complete Digital Operating System for University-Level Athletics Meets

The Problem

Today, a VTU athletics meet involves:

- Hundreds or thousands of athletes
- Coaches from dozens of colleges
- Multiple referees
- Multiple simultaneous events
- Accommodation allocation
- Food distribution
- Attendance verification
- Result processing
- Announcement management
- Photo and video sharing

Most of this is managed through:

- Paper sheets
- Printed schedules
- Phone calls
- WhatsApp messages
- Manual result compilation
- Physical notice boards

This leads to:

- Lost information
 - Delayed results
 - Attendance mistakes
 - Food misuse
 - Accommodation confusion
 - Last-minute communication failures
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Our Vision

Instead of managing a sports meet through 20 disconnected processes, VTU SportsOS provides a single unified digital platform where every stakeholder operates within the same ecosystem.

From the moment VTU uploads athlete registrations until the final medal table is published, the entire athletics meet is managed digitally.

User Roles

1. Athlete / Coach Portal

Used by:

- Athletes
 - Coaches
 - Team Managers
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2. Official Portal

Used by:

- Referees
 - Event Judges
 - Technical Officials
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3. Organizer Portal

Used by:

- Sports Committee
 - Accommodation Team
 - Food Committee
 - Event Coordinators
 - Administrative Staff
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Complete Event Lifecycle

Phase 1: Pre-Event Setup

Admin Dashboard

Event Control Center

The central administration hub for configuring and managing the athletics meet.

Functions

- Create event
- Set event dates
- Configure venues
- Configure accommodation blocks
- Configure food counters
- Upload schedules
- Create officials
- Manage permissions

Dashboard Metrics

Metric	Description
Total Athletes	Registered participants
Total Colleges	Participating colleges
Total Officials	Referees and staff
Rooms Occupied	Accommodation utilization
Meals Planned	Food planning statistics
Events Scheduled	Total events configured

Athlete Import System

VTU provides registration data containing:

- College
- USN
- Athlete Name
- Gender
- Registered Events

Automated Processing

Upon spreadsheet upload, the system automatically:

- Creates athlete profiles
- Creates coach profiles

- Maps athletes to events
- Generates participant IDs
- Allocates access credentials

Accommodation Management

Administrators configure:

Accommodation Locations

- Hostel A
- Hostel B
- Guest House
- Hotel Blocks

Room Configuration

Each room contains:

- Room Number
- Capacity
- Gender Restriction
- Current Occupancy

Automated Allocation Example

MIT Mysore Team

Accommodation	Rooms
Hostel A - Block C	Room 301
Hostel A - Block C	Room 302

Phase 2: Arrival Management

Athlete Home Page

When an athlete logs in, the dashboard displays:

- Accommodation Assignment
- Venue Map
- Event Schedule
- Digital Food Pass
- Notifications

Accommodation Details Page

Displays:

- Building Name
- Room Number
- Contact Person
- Google Maps Location

Actions

- Start Journey
 - I'm On Transit
 - I've Arrived
-

Arrival Tracking

Athlete Statuses

- Not Started
- On Transit
- Arrived
- Checked In

Organizer Dashboard

College-wise arrival monitoring.

Example:

College	Total Athletes	Arrived	In Transit
PES University	40	32	8

Phase 3: Digital Identity System

Chest Number Generation

Automatically generates unique identifiers.

Examples

Men

- M1001
- M1002
- M1003

Women

- W1001
 - W1002
-

Digital Athlete ID Card

Contains:

- Athlete Photo
- Name
- College
- Registered Events
- Chest Number
- QR Code

Single QR Usage

The same QR is used for:

- Check-in
 - Food distribution
 - Call room attendance
 - Event verification
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Phase 4: Food Management System

Athlete Food Pass

Displays meal eligibility.

Example

Day 1

- Breakfast
- Lunch
- Dinner

Day 2

- Breakfast
- Lunch
- Dinner

Meal Status

- Available
 - Consumed
 - Expired
-

Food Counter Operations

Volunteer scans athlete QR code.

System instantly verifies:

- Athlete Name
- Meal Type
- Eligibility

Example

Breakfast scanned:

Meal	Status	Time
Breakfast	Consumed	08:12 AM

Later:

Meal	Status
Breakfast	Expired
Lunch	Available

Benefits

- No printed coupons
 - No duplicate usage
 - Real-time food analytics
 - Consumption tracking
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Phase 5: Event Management

Athlete Event Dashboard

Displays:

- Event Name
- Date
- Time
- Venue
- Call Room Timing
- Reporting Countdown

Example

Event	Start Time
100m Sprint	10:30 AM

Reporting Deadline
10:00 AM

Phase 6: Smart Notification Engine

Automated notifications include:

24 Hours Before

Tomorrow:

- 100m Sprint

1 Hour Before

- Report to Call Room

30 Minutes Before

- Attendance Begins

Event Delays

Example:

100m Men's Sprint shifted to 11:00 AM

Only affected athletes receive notifications.

Phase 7: Digital Call Room

Current Method

Manual attendance and name calling.

SportsOS Method

Athlete enters call room.

Official scans QR code.

System records:

- Name
- Chest Number
- Event
- Timestamp

Attendance Status

- Present
- Late Reporting
- Absent

Automatic flag generation for late arrivals.

Phase 8: Heat and Lane Management

Applicable For:

- 100m
 - 200m
 - 400m
 - Relay Events
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Heat Creation

Automatically generates:

- Heat 1
 - Heat 2
 - Heat 3
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Lane Allocation Example

Lane	Athlete
1	M1024
2	M1030
3	M1041

Dynamic Recasting

If organizers reshuffle heats:

- Heat assignments update automatically
 - Athlete notifications are triggered instantly
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Phase 9: Referee Operations

Referee Dashboard

Each referee only sees assigned events.

Example:

100m Men

- Heat 1
 - Heat 2
 - Heat 3
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Result Entry Interface

Referees enter:

- Position
- Chest Number
- Timing

Example

Position	Chest No	Timing
1st	M1024	10.82

Validation Engine

Checks:

- Duplicate entries
 - Missing athletes
 - Lane inconsistencies
 - Invalid timings
-

Field Events

Supported Events:

- Shot Put
- Javelin Throw
- Discus Throw
- Long Jump
- Triple Jump

Tracking

Attempt 1	Attempt 2	Attempt 3
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Automatically calculates:

- Best Attempt
 - Ranking
 - Qualification
-

High Jump / Pole Vault

Tracks:

- Height Cleared
 - Number of Attempts
 - Failed Attempts
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Decathlon / Heptathlon

Tracks every stage individually.

Example

1. 100m Sprint
2. Long Jump
3. Shot Put

Automatically computes cumulative points.

Phase 10: Live Results System

The flagship feature of VTU SportsOS.

The moment a referee submits results:

- Results are published instantly
 - Rankings update automatically
 - Qualification status updates
 - Medal tables refresh in real time
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Athlete Result View

Example:

Event	100m Men
Position	2
Timing	10.91
Qualified	Yes

Live Medal Table

Displays:

- Gold
- Silver
- Bronze
- Total Points

Example

College	Gold	Silver	Bronze
MIT Mysore	5	2	4

Phase 11: Media Management

Media Team Portal

Official photographers upload:

- Photos
- Videos

Media stored through Google Drive integration.

Categorization

Media is organized by:

- Day
- Event
- Gender
- College

Example

Day 2

100m Men Final

- View Photos
 - View Videos
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Athlete Search

Athletes can search:

- Event Name
- College
- Day

To instantly access event media.

Phase 12: Command Center

The central operations dashboard displayed on large screens.

Provides real-time analytics across the entire event.

Arrival Analytics

Metric	Count
Registered Athletes	1500
Arrived	1450
In Transit	50

Food Analytics

Meal	Served
Breakfast	1320
Lunch	0
Dinner	0

Event Analytics

Metric	Count
Completed Events	17
Running Events	4
Upcoming Events	12

Result Analytics

Metric	Count
Pending Results	2
Published Results	56

Accommodation Analytics

Location	Occupancy
Hostel A	98%
Hostel B	82%

Phase 13: Closing Ceremony

Upon completion of the event:

Automatic Outputs

- Final Rankings
 - College Points Table
 - Medal Table
 - Certificates
 - Participation Records
 - Media Archive
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Final Outcome

VTU SportsOS transforms a traditional athletics meet from a collection of disconnected manual processes into a fully integrated digital operating system.

Benefits

Athletes

- Better visibility
- Instant results
- Easy navigation

Officials

- Faster operations
- Reduced paperwork
- Accurate records

Organizers

- Centralized management

- Real-time analytics
- Reduced operational overhead

VTU

- Professional event execution
 - Transparent result processing
 - Complete digital audit trail
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VTU SportsOS

One Platform. One Event. Complete Control.